

Personalized AI assistants could either promote or hinder the emergence of collective intelligence

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Abstract

Advances in artificial intelligence (AI) have enabled the wide use of personalized assistance for cognitive tasks, which enhanced individual performance on various fields. However, the impact of AI assistance on collective intelligence remains poorly understood. AI assistance may undermine the emergence of collective intelligence by reducing the individual diversity or introducing structured biases into a population. Here, we develop a human-AI collective prediction model mediated by social learning based on incentive structure by extending the existing works on collective intelligence. Our model uncovers that emergence of collective intelligence is depend on how human and AI interact. When AI works as if it is omniscient and gives assistance regardless of users' interests, collective intelligence will be promoted. However, when AI works as a chat-bot and gives assistance only based on users' interests, collective intelligence could be hindered due to diversity loss and overconfidence on AI. We showed that, to promote collective intelligence, penalizing or incentivizing AI usage is not reliable. Instead, elaborating more diversity-aware incentives could be the key to promote and maintain the collective intelligence.

Introduction

Collective intelligence (CI) describes a phenomenon in which a group’s ability to solve problems, make decisions, or adapt to changing environments exceeds each member’s ability [1, 2, 3, 4, 5]. This phenomenon, also referred to as wisdom of crowds, has thus served as a central mechanism underlying the success of diverse organizations, ranging from animal groups [6, 7, 8] and human societies [2, 9] to machine-learning systems [10, 11]. Understanding the conditions that promote and maintain CI therefore remains a fundamental question across social sciences [12], evolutionary biology [13], and computer science [14]. Addressing this question has become particularly timely as artificial intelligence (AI) has begun to reshape how individuals acquire information, make decisions, and contribute to collective outcomes [15].

The recent emergence of generative AIs and large language models (LLMs) has changed the way individuals engage in various cognitive tasks, ranging from simple everyday tasks to more complex tasks, including scientific discovery [16] and creative writing [17]. In particular, while the use of personal AI models, such as ChatGPT, has been rapidly increasing [18], the potential long-term impact of personal AI services on human society, especially the CI, remains uncertain. Some argue that AI can promote CI by enhancing individual expertise [19], and opinion aggregation [20]. Conversely, empirical evidence indicate that the use of AIs can limit the diversity of opinions [17, 21, 22], a key requirement for the emergence of CI [23, 24, 25].

Our ability to predict the long-term impact of personal AI services on human CI is hampered by the apparent discrepancy in the incentives that shape AI uses. On one hand, many industrial sectors are pushing towards increased AI usages, where the number of “tokens” used is often perceived as a proxy for productivity [26, 27]. On the other hand, many academic institutions appear to discourage, and even penalize, AI use, reflecting concerns about plagiarism, misinformation, and limited independence [28, 29]. Such discrepancy raises a question about whether directly incentivizing or penalizing AI usage will benefit or harm human CI. These observations further highlight the importance of indentifying a robust incentive structure that can promote CI in the presence of AI, which can retain the benefits of AIs while preventing potential risk such as preserving the diversity of individual opinions.

To address these questions, we extended previous approaches for modeling CI, where each individual makes independent predictions for the outcome from partial observations of underlying causal factors [12, 13, 30]. We allowed each individual to independently consult with personal AIs and revise their predictions accordingly. Across different assumptions about the AI model’s ability and underlying incentive structures, we investigate the robustness of human CIs.

Main

Model

We begin by extending previous models of CI in which a group of players makes collective predictions by gathering information and aggregating individual predictions in a decentralized manner [30, 12, 13]. In the previous approach, among M different factors (x_i) that affect the outcome (Y), an individual (A) makes a prediction based on their factor of interest (x_{e_A}) and their own belief (c_A): $Y_A = c_A x_{e_A}$. Here, we assume each individual can independently consult AI and revise their prediction based on their reliance on AI (β_A): $Y_A^{re} = \beta_A Y_A^{AI} + (1 - \beta_A) Y_A$.

To specify how individuals consult AI, we model the AI prediction (Y_A^{AI}) as a noisy and biased estimate of the target outcome. Specifically, AI predictions include factor-specific bias (b_i) and stochastic error (ϵ). We then consider two extreme modes of AI response, which differ in the amount of information AI uses when consulting an individual user. The first is the “AI knows all” model, in which AI predicts the outcome based on all factors regardless of the user’s interest: $Y_A^{AI} = \sum_{i=0}^M (\alpha_i x_i + b_i) + \epsilon$, where α_i represents the true coefficient of the factor. The second is the “AI answers question” model, in which AI predicts the outcome based only on the user’s interest: $Y_A^{AI} = \alpha_{e_A} x_{e_A} + b_{e_A} + \epsilon$.

After the prediction, each individual is rewarded by the expected value of their payoff under the incentive structure. We adopt two incentive structures, “feedback” and “niche expert” from Wang et al. [13], which have been shown to promote near-perfect collective accuracy without AI (except averaging aggregation with niche expert incentive). The feedback incentive gives higher payoff to individuals who can push the collective prediction toward the true outcome, regardless of their individual predictive accuracy. By contrast, the niche expert incentive gives higher payoff to individuals who have both better predictive accuracy and a minority interest.

Through time, the group evolves by imitating the peers who got better payoff. Specifically, for each generation, two individuals are randomly sampled, and one imitates all three strategies (factor of interest, belief, reliance on AI) of the other with a probability determined by the difference in expected payoff. Individual predictions are aggregated into a collective prediction by simple averaging in the AI knows all model, or by clustering in the AI answers question model, depending on the mode of AI.

AI can either promote or prevent the emergence of CI

To understand how AI could affect the emergence of CI, we first compare collective dynamics with and without AI assistance. We find that AI affects CI in qualitatively different ways depending on the mode of AI.

In the AI knows all model, AI promotes CI by increasing either the speed at which CI emerges or the resulting collective accuracy (Fig. 2A). When the group can

103 achieve near-perfect CI without AI (feedback incentive), AI accelerates its emergence
 104 by raising the initial collective accuracy to the level of AI accuracy. By contrast,
 105 when the group cannot achieve CI without AI (niche expert incentive), AI enables
 106 moderately accurate CI that exceeds the accuracy achievable by either AI or the
 107 human group alone. These trends are robust across different levels of AI accuracy
 108 and can be shown mathematically by assuming a sufficiently large group of players
 109 (see SI).

110 In contrast, In the AI answers question model, AI prevents the emergence and
 111 maintenance of CI (Fig. 2B). Under the feedback incentive, the group initially
 112 achieves near-perfect CI. However, as interest diversity rapidly decreases, the group
 113 fails to capture variation in some factors, leading to increased collective variance
 114 and a breakdown of collective accuracy. Under the niche expert incentive, collective
 115 accuracy initially increases but peaks before near-perfect CI is achieved, and then
 116 plateaus with a slight decline. This trend is mediated by increased reliance on AI:
 117 individuals with higher reliance on AI obtain higher payoffs, causing the group to
 118 become increasingly reliant on AI and the collective prediction to inherit AI bias.
 119 However, the niche expert incentive can also drive reliance on AI toward zero, de-
 120 pending on the relationship between the true coefficient and AI bias. In this case,
 121 the group excludes AI assistance and builds CI on its own, which largely delays the
 122 emergence of CI (see SI).

123 **Directly incentivizing or penalizing AI use can harm CI**

124 Given the strong influence of AI use on collective outcomes, a straightforward inter-
 125 vention is to directly incentivize or penalize AI use itself. To investigate whether such
 126 direct incentives or penalties improve CI, we modify the two incentive structures by
 127 adding a payoff term proportional to individual reliance on AI:

$$\text{Feedback/Niche expert} + \lambda\beta_A \quad (1)$$

128 where λ represent the strength of incentive ($\lambda > 0$) or penalty ($\lambda < 0$). Except in the
 129 AI knows all model under the feedback incentive, directly incentivizing or penalizing
 130 AI use fails to promote near-perfect CI. Moreover, in most cases, directly incen-
 131 tivizing or penalizing leads to lower collective accuracy than the original incentive
 132 structures (Fig. 3).

133 When AI use is directly incentivized, reliance on AI evolves toward full reliance
 134 ($\beta_A = 1$), reducing collective accuracy as the collective prediction inherits AI bias.
 135 By contrast, when AI use is directly penalized, group evolves toward no reliance on
 136 AI ($\beta_A = 0$), where collective accuracy decreases with the strength of the penalty. In
 137 some cases, when AI bias is relatively large and the penalty is weak, penalizing AI use
 138 can achieve higher collective accuracy than the original incentive structure. However,
 139 in most cases, penalizing AI use imposes strong selection against AI reliance, pushing
 140 the group toward lower stationary collective accuracy. Although the AI answers
 141 question model under the feedback incentive already shows stochastic breakdown of

142 CI due to the loss of interest diversity, incentivizing or penalizing AI use further
143 strengthens the breakdown.

144 **Balanced incentives mitigate AI-induced failures of CI**

145 Focusing on the AI answers question model, in which both incentive structures fail
146 to promote accurate CI, we investigate whether a better incentive structure can over-
147 come the risks introduced by AI. Here, we propose a “balanced” incentive, defined
148 as the simple average of the feedback and niche expert incentives.

149 The core idea of the balanced incentive is to maintain both within- and between-
150 interest diversity. The feedback incentive can strongly maintain within-interest di-
151 versity, defined as diversity of strategy (belief, reliance on AI) among individuals with
152 the same interest. However, between-interest diversity, which we denoted as interest
153 diversity in the previous sections, can be hindered when AI assistance is present.
154 This occurs because revising individual predictions with AI predictions compresses
155 the belief distribution of the group. Although the initial belief distribution is shifted
156 toward the true value, the compressed distribution makes it harder for the group to
157 achieve near-perfect predictions through social learning for important factors (factors
158 with large absolute coefficients). This eventually leads to a strong flow of players
159 from relatively less important factors to more important factors, causing a loss of
160 between-interest diversity.

161 By contrast, the niche expert incentive can maintain between-interest diversity,
162 because it directly incentivizes minority interests. However, it cannot maintain
163 within-interest diversity because, within the same interest, individuals are rewarded
164 only according to their individual predictive performance. This leads to strong se-
165 lection for either high or low reliance on AI, which in turn results in lower collective
166 accuracy.

167 The balanced incentive can promote near-perfect CI by maintaining both within-
168 and between-interest diversity, regardless of AI accuracy. Reliance on AI decreases as
169 AI accuracy decreases, reflecting the effective use of AI assistance. However, human
170 accuracy, defined as collective accuracy without AI, remains largely stochastic, and
171 only moderate accuracy can be achieved. This suggests that, although an appropriate
172 incentive structure can promote robust CI under AI assistance, the group may need
173 to rebuild CI through further adaptation when AI assistance is inaccessible.

174 **Discussion**

175 We present an epidemiological analysis of RSV outbreaks in Japan combining spa-
176 tiotemporal observations with dynamical disease modeling. Our analysis revealed
177 semiannual cycles in seasonal RSV transmission in four major islands (Honshu,
178 Shikoku, Kyushu, and Ryukyu), which correlate with specific humidity and tem-
179 perature. We found that these semiannual cycles allowed a sudden shift in the
180 seasonality of RSV outbreaks in response to an increase in RSV transmission—this

181 increase could be captured by either through changes in susceptibility or transmissi-
182 bility. We hypothesize that an increase in childcare capacity through Comprehensive
183 Support System for Children and Childcare may have contributed to the increase in
184 RSV transmission [?].

185 Our analysis revealed considerable heterogeneity in epidemic dynamics across
186 major islands in Japan. We showed that these differences could be explained by
187 the differences in underlying seasonal transmission. Notably, we found a robust,
188 quadratic relationship between the estimated transmission rates against mean spe-
189 cific humidity and mean temperature across four major islands (Honshu, Shikoku,
190 Kyushu, and Ryukyu), indicating low transmission at intermediate levels of specific
191 humidity. These findings echo earlier studies that demonstrated similar relationships
192 for RSV [?] and influenza [? ? ? ? ?]. However, given strong correlation between
193 specific humidity and temperature, we were unable to identify the primary environ-
194 mental driver from the current study. Their relative contributions may be separated
195 from data spanning a wider geographical region with more climate variability as well
196 as epidemiological variability [?].

197 The robustness of this quadratic relationship across islands exhibiting different
198 climate conditions also suggests a possibility that climate-driven transmission may
199 be, in part, facilitated by human behavior: for example, an increase in time spent
200 indoors during low and high humidity seasons can contribute to increased transmis-
201 sion. Further investigation is needed to establish the underlying mechanism behind
202 how climate conditions affect the transmission of respiratory pathogens, especially
203 across different environmental contexts (e.g., indoor vs outdoor). We note that this
204 relationship is correlational, rather than causal, and therefore any other climate vari-
205 ables (e.g., rainfall) or seasonal variation in human behavior (e.g., school terms) that
206 correlate with seasonal variation in specific humidity and temperature will be implic-
207 itly captured by this relationship. Understanding why the relationship between RSV
208 transmission and humidity in Hokkaido island differs from other locations remains
209 to be answered.

210 We hypothesized that the Comprehensive Support System for Children and Child-
211 care may have contributed to the increase in RSV transmission, but other mecha-
212 nisms may have also contributed. For example, an increase in inbound overseas trav-
213 elers may have led to an increased exposure to RSV among adults thereby increasing
214 the total amount of transmission [?]. However, RSV typically infects young chil-
215 dren [?] and contact rates among young children and adolescents (5–19) are much
216 higher than among older adults in Japan [?], suggesting that an increase in
217 inbound overseas travelers may have smaller effects than an increase in childcare at-
218 tendance on overall RSV dynamics. Another competing hypothesis that could lead
219 to an increase in transmission would be strain evolution: for example, one study
220 noted that L172Q/S173L mutant strains of RSV B that became dominant around
221 2016 had reduced susceptibility against monoclonal antibodies [?]. However, it is
222 not yet clear how this mutation translates to susceptibility against infection-derived
223 antibodies. While our findings are consistent with those by [?], who also pointed

out the association between an increase in childcare attendance and a large change in the seasonality of Kawasaki disease in Japan, we cannot rule out the possibility that other factors, such as increase in air travel [?] or strain evolution [?], could have contributed to the increase in overall transmission.

Our sensitivity analysis revealed that an increase in transmissibility, rather than an increase in susceptibility, can also explain a sudden shift in seasonality in RSV outbreaks. In fact, both models provide evidence for increased contact rates. Specifically, the total rate of RSV transmission can be written as $\hat{\beta}cSI/N$, where $\hat{\beta}$ represents the rate of transmission per contact, c represents the contact rate, S represents the number of susceptible individuals, I represents the number of infected individuals, and N represents the total population size. Then, an increase in contact rate c can be captured by either increase in S (the original model) and $\hat{\beta}$ (the sensitivity analysis). Therefore, conclusions from both models are consistent with our hypothesis that an increase in contact rates, especially among children, contributed to the sudden change in RSV seasonality. Detailed age structured surveillance data, alongside age structured model, will be needed to properly assess the relative contribution of potential mechanisms, such as increase in childhood mixing patterns or increase in inbound overseas travelers.

Our model-based estimate of the initial susceptible fraction range between 8%–11%. However this estimate must be interpreted with care as it represents effective changes in population-level susceptibility, which can depend on many factors such as immunological and behavioral changes, rather than an actual susceptible fraction. Therefore, these estimates are not directly comparable to serological estimates of proportion seronegative. For example, [?] recently estimated that 90% of individuals above 3 years of age have been infected at least once by analyzing anti-pre-F protein antibody concentration data from the Netherlands [? ?], but individuals with RSV antibodies can still permit reinfection. A cross-section serological assay from healthcare and non-healthcare workers at a pediatric medical facility in Japan found that 95% of the participants were seropositive but only 63.5% had greater than 8 fold neutralizing antibody titers [?], suggesting challenges in translating serological data to susceptible fraction. While our estimates are not inconsistent with these data, detailed age structured serological data and model are needed to accurately estimate changes in population-level susceptibility against RSV.

Interventions to slow the transmission of COVID-19 have disrupted the circulation of many pathogens [? ? ? ?], including RSV epidemics in Japan. This disruption has added major challenges to predicting future outbreaks, which prevented us from making long-term predictions. Continued analysis of RSV dynamics in the post-COVID period, particularly with regard to whether RSV outbreaks in Japan return to fall or winter outbreaks, may help further validate our models.

Our analysis relied on several simplifying assumptions. For example, our model assumed that the waning of immunity can render previously infected individual to become fully susceptible to reinfections; this approximation allowed us to reconstruct the dynamics of susceptible hosts more easily. In practice, immunity is likely more

complex with secondary infections being less susceptible and transmissible than primary infections [?]. Other studies have also suggested the importance of interaction between RSV A and B [? ?] as well as competition between RSV and human metapneumovirus [?]; our model did not account for such strain dynamics. We also did not account for explicit spatial structure or underlying stochasticity of the system. Therefore, our estimates of transmission rates must be interpreted with caution as they may implicitly capture factors that we did not account for explicitly, including strain dynamics, spatial structure, and exogenous transmission. Our analyses also primarily focused on three major islands (Honshu, Shikoku, and Kyushu), necessitating a better understanding of RSV dynamics in Hokkaido and Ryukyu islands; however, we note that these three islands make up $> 95\%$ of the population in Japan, meaning that our analyses capture the majority of RSV transmission in Japan. Despite these limitations, our model likely represents a parsimonious approximation of the complex host-pathogen interactions, allowing us to draw general conclusions about how interactions between endogenous (contact rates) and exogenous (climate-driven factors) factors can give rise to a sudden dynamical transition.

Understanding how endogenous and exogenous factors shape epidemic dynamics is critical to predicting future outbreaks and making public health decisions. Our analysis shows that the interplay between climate-driven transmission and subtle changes in contact rates can cause a sudden transition in epidemic dynamics. More broadly, our analysis demonstrates that detailed epidemiological time series data can allow us to tease apart endogenous and exogenous factors in explaining dynamical transitions, offering unique insights into a long-standing ecological question.

Methods

To model CI, we start with an abstract model that assumes the prediction target is the result of multiple causal factors [13]. Specifically, the prediction target Y is modeled as a linear combination of M independent factors $(x_1, x_2, \dots, x_m)$ and $M + 1$ coefficients $(\alpha_0, \alpha_1, \dots, \alpha_m)$, where α_0 denotes the global offset (Fig. 1A). The M factors are randomly drawn from normal distributions with mean X_i and standard deviation σ_i in each generation.

Among a population of N individuals, each individual can observe a single factor and make a prediction based on the observed value and their own belief. In particular, individual A 's prediction is modeled as:

$$Y_A = c_A x_{e_A} \quad (2)$$

where c_A represents individual A 's personal belief, e_A represents the factor that A observes (interest), and x_{e_A} represents the realized value of A 's interest. Individuals can also choose not to observe any factor, which represents stubborn individuals who make predictions only based on their own belief. For notational convenience, we assume that stubborn individuals have interest 0, and let x_0 always be 1.

Next, we model AI as a prediction system that has learned the true coefficients with bias and error (Fig. 1B). For each factor $i \in [0, M]$, AI knows the exact coefficient α_i and has a fixed bias b_i , resulting in a prediction of $\alpha_i x_i + b_i$. In terms of human-AI interaction, we model two contrasting modes of AI: the “AI knows all” model and the “AI answers question” model. In the AI knows all model, AI acts as an omniscient system and gives an answer regardless of the individual’s interest, accounting for all factors related to the task. Thus, AI predicts as follows:

$$Y_A^{AI} = \sum_{i=0}^M (\alpha_i x_i + b_i) + \epsilon \quad (3)$$

In contrast, the AI answers question model acts as a chatbot and gives an answer only based on the individual’s interest. Thus, given the individual’s interest, e_A , AI predicts as follows:

$$Y_A^{AI} = \alpha_{e_A} x_{e_A} + b_{e_A} + \epsilon \quad (4)$$

In both models, $\epsilon \sim N(0, \sigma_e^2)$ represents random error arising from the stochastic generative process of the AI model.

AI works as a personalized assistant for each individual, such that each individual receives assistance by incorporating the AI’s prediction. However, even with the same opportunity to receive AI assistance, reliance on AI can vary substantially across individuals. To model this, we assume that the revised prediction is a weighted sum of the individual’s own prediction and the AI’s prediction by introducing AI belief β_A as a weighting coefficient. The revised prediction of individual A can be written as:

$$Y_A^{revised} = \beta_A Y_A^{AI} + (1 - \beta_A) Y_A \quad (5)$$

The revised predictions of individuals are aggregated by two different mechanisms, depending on the mode of AI. In the AI knows all model, even if an individual can observe only a single factor, their revised prediction contains information about all factors through interaction with AI. Therefore, we use simple averaging of revised predictions as the collective prediction. In contrast, in the AI answers question model, each individual gives a prediction based only on a single factor even after revising their prediction with AI assistance. Therefore, we use clustering aggregation to obtain the collective prediction, which is performed by clustering individuals based on interest and summing the average values of each cluster.

For each generation, each individual is rewarded with a payoff based on the given incentive structure (Fig. 1C). We adopt two incentive structures, “feedback” and “niche expert,” which were proposed by Wang et al. [13] and promote collective intelligence when prediction is performed without AI. The feedback incentive gives higher payoff to individuals who can push the collective prediction toward the true value, regardless of their individual performance. Given individual A ’s revised prediction $Y_A^{revised}$ and the collective prediction $\hat{Y}^{revised}$, it can be expressed as:

$$\pi_A = Y_A^{revised}(Y - \hat{Y}^{revised}) \quad (6)$$

340 The niche expert incentive gives higher payoff to individuals who have both better
 341 individual performance and a minority interest. Specifically, the niche expert payoff
 342 is expressed as the product of the proportion of individuals who observe the same
 343 factor as individual A , ρ_{e_A} , and individual accuracy. Given individual A 's revised
 344 prediction $Y_A^{revised}$, it can be expressed as:

$$\pi_A = -\rho_{e_A}(Y_A^{revised} - Y)^2 \quad (7)$$

345 These two incentive structures were shown to achieve nearly perfect collective in-
 346 telligence through social learning (except for the niche expert incentive with simple
 347 averaging aggregation) by promoting individual expertise while maintaining collec-
 348 tive diversity. Therefore, we use these incentive structures as a baseline to under-
 349 stand how AI affects the emergence of collective intelligence. Moreover, we modify
 350 the two baseline incentives in a later section to understand the consequences of direct
 351 incentives and penalties for AI usage.

352 Throughout the simulation, individuals engage in social learning by imitating
 353 other individuals who receive higher payoffs. We assume that prediction events
 354 occur more frequently than social learning events within the population, such that
 355 individual payoff can be computed as an expected payoff. The imitation process
 356 is performed for at most one pair in each generation. By sampling two random
 357 individuals, A and B , imitation is performed with probability:

$$\frac{1}{1 + \exp(s(\mathbb{E}[\pi_A] - \mathbb{E}[\pi_B]))} \quad (8)$$

358 where s denotes the strength of social learning. When imitation occurs, A imitates
 359 all three strategies of B : interest, belief, and AI belief.

360 To evaluate and analyze the dynamics of collective prediction, we define six sum-
 361 mary metrics: collective accuracy, collective bias, collective variance, human accu-
 362 racy, interest diversity, and median AI belief. These metrics capture the accuracy
 363 of the revised collective prediction, its bias-variance decomposition, the predictive
 364 performance without AI, the diversity of individual interests, and median AI belief
 365 in the population (Table 1).

366 Simulations

367 We run a series of simulations to understand how the interplay between climate-
 368 driven transmission and population-level susceptibility can drive a sudden shift in
 369 the timing of an epidemic. First, we simulated the model for a year (from week 26 of
 370 the starting year to week 25 of the following year) by varying the initial conditions
 371 and computing the center of gravity. Specifically, we varied $i(0)$ between 1×10^{-4}
 372 and 3×10^{-3} and $s(0)$ between 0.078 and 0.105. All other parameters were set to
 373 posterior median estimates.

Table 1: Evaluation metrics used to analyze collective prediction dynamics.

Metric	Definition	Interpretation
Collective accuracy	$1 - \frac{\mathbb{E}[(Y - \hat{Y}^{\text{revised}})^2]}{\mathbb{E}[(Y - \alpha_0)^2]}$	Accuracy of the revised collective prediction relative to the baseline prediction α_0 .
Collective bias	$\left(\mathbb{E}[Y - \hat{Y}^{\text{revised}}]\right)^2$	Squared systematic deviation of the revised collective prediction from the true value.
Collective variance	$\text{Var}(Y - \hat{Y}^{\text{revised}})$	Variability of the revised collective prediction error across prediction events.
Human accuracy	$1 - \frac{\mathbb{E}[(Y - \hat{Y})^2]}{\mathbb{E}[(Y - \alpha_0)^2]}$	Accuracy of the human-only collective prediction relative to the baseline prediction α_0 .
Interest diversity	$\exp\left(-\sum_{i=0}^M \rho_i \ln \rho_i\right)$	Effective number of interests represented in the population.
Median AI belief	$\text{median}_{A \in N}(\beta_A)$	Median value of AI belief β_A in the population.

374 To further understand how the shape of seasonal transmission term affects the
 375 degree of shift in seasonality, we varied the shape of seasonal transmission term by
 376 interpolating the estimated β_{seas} from Honshu and Kyshu islands, which are the two
 377 most populated islands. To do so, we first took posterior median estimates of β_{seas}
 378 from two islands and fitted generalized additive model with cyclic cubic spline bases
 379 to obtain smoothed estimates of β_{seas} for each island, which we denote as β_H and
 380 β_K , respectively. Then, we normalized seasonal transmission rates such that it has
 381 a mean of zero and has an amplitude of 1:

$$\zeta_X(t) = \frac{1}{\alpha_X} \left(\frac{\beta_X(t)}{\beta_X} - 1 \right) \quad (9)$$

382 where $\zeta_X(t)$ represents the normalized seasonal transmission pattern in island X , and
 383 $\alpha_X = (\max(\beta_X(t)) - \min(\beta_X(t)))/2$ represents the amplitude of seasonal transmission
 384 pattern in island X . This allowed us to interpolate between two normalized seasonal
 385 terms and obtain a flexible shape for seasonal transmission rate:

$$\zeta_{\text{new}}(t) = \zeta_H(t)(1 - \theta) + \zeta_K(t)(\theta), \quad (10)$$

$$\beta_{\text{new}} = \left(\frac{\alpha_{\text{new}} \zeta_{\text{new}}(t)}{(\max(\zeta_{\text{new}}(t)) - \min(\zeta_{\text{new}}(t)))/2} + 1 \right) \bar{\beta}_{\text{new}}, \quad (11)$$

386 where θ represents the interpolation coefficient, such that $\theta = 0$ and $\theta = 1$ causes
 387 β_{new} to have the same shape as β_H and β_K , respectively and $0 < \theta < 1$ allows
 388 us to model counterfactual transmission scenarios that interpolates between two
 389 islands. Note that $\zeta_{\text{new}}(t)$ does not necessarily have an amplitude 1 so we divide
 390 it by $(\max(\zeta_{\text{new}}(t)) - \min(\zeta_{\text{new}}(t)))/2$ to ensure the amplitude of 1.

391 For a given value of the interpolation coefficient θ and seasonal amplitude α_{new} ,
 392 we simulated two outbreaks for a year assuming $s(0) = 0.078$ and $s(0) = 0.105$
 393 and computed the difference in the timing of epidemic peak. In doing so, all other
 394 parameters, including the mean transmission rate β_{new} , were fixed to posterior median
 395 estimates for the Honshu island.

396 Regression

397 We performed univariate quadratic regressions for the estimated transmission rates
 398 against mean specific humidity and mean temperature, separately, and computed R
 399 squared values for each regression. We also perform bivariate quadratic regressions
 400 for the estimated transmission rates using both mean specific humidity and mean
 401 temperature as covariates. The significance of regression coefficients was determined
 402 using two-sided t-tests at the 0.05 significance level.

403 We performed a linear regression between the estimated proportion of susceptibles
 404 at the beginning of each season (week 26) against the number of children attend-
 405 ing childcare facilities. For simplicity, the regression was performed using median
 406 estimates for the susceptible proportions. We did not have data on the number of
 407 children attending childcare facilities broken down by island level and so we used the
 408 national-level data instead.

409 Data availability

410 All data are stored in a publicly available GitHub repository ([https://github.com/](https://github.com/parksw3/perturbation)
411 [parksw3/perturbation](https://github.com/parksw3/perturbation)) [?].

412 Code availability

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518 Contributions

519 S.W.P., I.H., and B.T.G. conceived of the study. S.W.P. performed the analysis and
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522 Competing Interests

523 The authors declare no competing interests.

524 Figures

Figure 1: **Observed changes in RSV outbreak dynamics in Japan.** (A) Relative RSV cases across 47 prefectures in Japan between 2013 and 2024. Relative cases, representing the relative magnitude of RSV outbreaks in each prefecture compared to RSV outbreak sizes before the COVID-19 pandemic, are calculated by dividing the raw cases by the maximum value before the COVID-19 pandemic. The red arrow indicates when the shift in seasonality occurred. (B) Estimates of center of gravity (i.e., the mean timing of an epidemic) across 46 prefectures, excluding Okinawa. (C) Estimates of the epidemic trough (i.e., the minimum number of cases in a season divided by the population size) across 47 prefectures. In panels B and C, the center (horizontal line), lower bounds, and upper bounds of the box plot correspond to median, lower quartile (25th percentile), and upper quartile (75th percentile), respectively. The whiskers indicate the range of values that extend up to 1.5 times the interquartile range beyond the lower and upper quartiles. Outliers, which fall outside this range, are plotted individually. (D) Time series of RSV cases across 5 major islands.

Figure 2: **Summary of SIRS model fits to RSV outbreaks across major islands in Japan.** (A) Comparisons of observed cases (points) across the five major islands and fitted epidemic trajectories (red lines). (B) Estimated periodic seasonal transmission rates. (C) Relationship between the estimated periodic seasonal transmission rates and mean specific humidity. Points represent seasonal transmission rate estimates across 52 weeks versus average humidity across 2013–2020. Blue lines and regions represent the corresponding locally estimated scatterplot smoothing (LOESS) estimates and corresponding 95% confidence intervals ($n=52$). Red lines and regions represent the corresponding quadratic regression fits and corresponding 95% confidence intervals ($n=52$). (D) Estimated relative changes in transmission, capturing the impact of NPI measures. (E) Estimated proportion of the susceptible pool. In panels B, D, and E, lines represent the estimated median of the posterior distribution ($n=8000$ posterior samples). In panels B, D, and E, shaded regions represent the 95% credible intervals from the posterior distribution ($n=8000$ posterior samples).

Figure 3: **An increase in the susceptible pool explains sudden changes in seasonality.** (A) Predicted effects of the proportion of infected $i(0)$ and susceptible $S(0)$ at the beginning of season on center of gravity. Points represent the estimated values for $i(0)$ and $s(0)$ between 2013 and 2019, showing the last two digits of a given year. The white vertical dashed line represents the $i(0)$ value used for simulating epidemic dynamics in panel B. (B) Changes in epidemic trajectories caused by an increase in the susceptible proportion at the beginning of season for a fixed value of $i(0)$. (C–F) Comparisons of interpolated transmission rates used for simulating the SIRS model, corresponding to each corner in Panel G. Black lines represent the transmission rates used for simulations. Gray lines represent the estimated transmission rates the Honshu island as a visual reference. (C) The estimated transmission rates for the Honshu island. (D) The resulting transmission rates for the Honshu island with equal amplitude as the Kyushu island. (E) The resulting transmission rates for the Kyushu island with equal amplitude as the Honshu island. (F) The estimated transmission rates for the Kyushu island. (G) Differences in peak epidemic timing when we increase the the susceptible proportion at the beginning of season from 7.8% to 10.5% using transmission patterns that interpolate the estimates for Honshu and Kyushu islands.

Figure 4: **Increase in the susceptible pool following the launch of the Comprehensive Support System for Children and Childcare in Japan.** (A) Direct comparisons between the estimates of susceptible proportion at the beginning of each season in each island (black) and the number of children attending childcare facilities in Japan (red). Points represent median from $n=8000$ posterior samples. Shaded regions represent the 95% credible interval in our estimates ($n=8000$ posterior samples). (B) Correlations between the estimates of susceptible proportion at the beginning of each season in each island and the number of children attending childcare facilities in Japan. Points represent median from $n=8000$ posterior samples. Error bars represent the 95% credible interval in our estimates ($n=8000$ posterior samples). Red lines and shaded regions represent the best fitting linear regression and the corresponding 95% confidence intervals.